

Vedic Astrology and Astronomical Features in Tropical Cyclone Intensity Modelling: A Large-Scale Empirical Assessment

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Abstract

We present the first large-scale empirical investigation of whether Vedic Panchang (Hindu calendar) categories and planetary astronomical positions carry detectable signal for tropical cyclone intensity. Using 47,533 track observations from 944 global storms spanning 2016–2026, we construct a 222-feature dataset encompassing 14 meteorological, 18 Panchangam, and 190 scientific-astronomical predictors, and evaluate predictive performance through multiple complementary methods: chronological holdout regression, five-fold forward-chaining cross-validation, permutation group importance, non-parametric hypothesis testing with effect size quantification, and confound-adjusted stratified analysis. Our primary finding is a **null result**: adding Vedic features to a meteorological baseline degrades wind-change root-mean-square error (RMSE) by 14.3% and the full model fails to outperform the meteorological baseline in 4 of 5 temporal folds. Two secondary signals survive Bonferroni correction but carry negligible practical effect sizes: a lunar-phase association with rapid intensification (RI; $\chi^2 = 19.66$, $p = 0.006$, Cramér’s $V \approx 0.020$) and a consistent negative correlation between planetary speed and storm bearing direction ($\rho \approx -0.18$, $r^2 \approx 3\%$). We document that statistical significance is inflated by dataset scale ($N \approx 47,533$) and identify seasonal confounding as the primary driver of apparent Vedic categorical effects. These findings provide a methodological template for rigorous evaluation of non-conventional features in meteorological machine learning.

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1 Introduction

Tropical cyclones pose severe societal risks, and improving the accuracy of intensity forecasts remains an active area of meteorological research [1]. Alongside conventional numerical weather prediction and machine learning approaches that exploit atmospheric and oceanic variables, there exists a long-standing traditional claim, rooted in Vedic astrology, that planetary positions and Hindu calendar cycles modulate weather events and natural disasters. Vedic astrology encodes the positions of the Sun, Moon, and seven classical planets into a rich system of named categories: *Tithi* (lunar day), *Nakshatra* (lunar mansion), *Yoga* (Sun–Moon arc interval), *Karana* (half-Tithi), and several dozen more. These categories are deterministic functions of planetary ephemerides and are therefore computable for any date, time, and location.

This determinism makes Vedic features unusual among non-conventional predictors: they are neither noisy survey data nor domain-expert judgements but precise mathematical transformations of astronomical coordinates. If any Vedic category were genuinely associated with cyclone intensity, it would imply that the underlying astronomical geometry carries information about storm dynamics beyond what standard meteorological variables encode.

To our knowledge, no prior study has subjected these claims to a rigorous large-scale test against a diverse global cyclone database. We address this gap by assembling data from 944 storms across all major ocean basins, pairing track observations with Vedic Panchang values computed using the Drik Panchang astronomy engine, and conducting a battery of tests whose design explicitly addresses the statistical pathologies common in large- N astronomical association studies.

Research Questions

- RQ1.** Do Vedic Panchang features improve next-step storm intensity prediction over a meteorological baseline?
- RQ2.** Is there a statistically and practically significant association between lunar phase and rapid intensification?
- RQ3.** Do planetary speeds correlate with storm track bearing beyond chance?
- RQ4.** Do planets in apparent retrograde motion correspond to measurably different storm intensity distributions?
- RQ5.** Do Vedic categorical wind associations persist after removing the dominant seasonal–geographic confound?

Summary of Contributions

- First large-scale ($N = 47,533$ observations) empirical test of Vedic astrology features in tropical cyclone forecasting.
- Comprehensive effect size analysis distinguishing statistical from practical significance across all findings.
- A confound-adjusted partial Kruskal–Wallis framework for testing categorical astronomical associations within season $\times$ basin strata.

- A novel lunar-phase $\times$ lifecycle interaction analysis showing that RI associations with lunar phase are lifecycle-stage dependent.
- Open release of the 222-feature parsed dataset and all analysis scripts.

2 Data and Features

2.1 Storm Track Data

We use best-track data assembled from multiple regional operational meteorological agencies for 944 tropical cyclones active between July 2016 and January 2026, spanning the North Atlantic, Western Pacific, North Indian Ocean, South Indian Ocean, and South Pacific basins. Each track observation records the storm centre position (latitude, longitude), maximum sustained wind speed (knots), minimum central pressure (hPa), and UTC timestamp. Track observations are provided at variable intervals (predominantly 3–6 h), yielding 47,533 observations in total after filtering incomplete records.

2.2 Vedic Panchang Features

For each track observation, Vedic Panchang attributes were computed using the Drik Panchang astronomy engine at the storm centre location and local time. This produces 18 categorical and derived features (Table 1), including the five canonical Panchang limbs plus additional calendar and celestial quantities. Continuous Vedic planetary parameters (sidereal degree, speed in degrees per day, right ascension, declination, and Nakshatra pada) were computed for 15 bodies: Sun, Moon, Mars, Mercury, Jupiter, Venus, Saturn, Uranus, Neptune, Rahu (north lunar node), Ketu (south lunar node), Yama, Arun, Varun, and Spashtha Rahu/Ketu, yielding 75 numeric Vedic features.

Table 1: Vedic Panchang categorical features used in this study.

Feature	Type	Description
Tithi	Categorical (30)	Lunar day (Sun–Moon elongation in 12° steps)
Nakshatra	Categorical (27)	Lunar mansion (Moon’s sidereal position in 13.3° segments)
Yoga	Categorical (27)	Sun+Moon sidereal longitude mod 360° , divided into 27 equal parts
Karana	Categorical (11)	Half-Tithi period
Paksha	Binary	Waxing (<i>Shukla</i>) or waning (<i>Krishna</i>) lunar fortnight
Moonsign	Categorical (12)	Sidereal zodiac sign of the Moon
Sunsign	Categorical (12)	Sidereal zodiac sign of the Sun
Drik Ritu	Categorical (6)	Observational season (equinox/solstice based)
Vedic Ritu	Categorical (6)	Traditional Hindu season
Drik Ayana	Binary	Northern/southern hemisphere Sun path
Vedic Ayana	Binary	Traditional equivalent of Ayana
Anandadi Yoga	Categorical (9)	Auspiciousness index derived from weekday+Nakshatra
Tamil Yoga	Categorical (5)	Tamil calendar auspiciousness classification
Samvatsara	Categorical (60)	60-year Jupiter cycle year name
Rahu Kalam	Time range	Daily inauspicious period (time-range string)
Yamaganda	Time range	Secondary inauspicious period
Gulikai Kalam	Time range	Tertiary inauspicious period
Weekday	Categorical (7)	Hindu weekday

2.3 Astronomical Features (Skyfield)

Skyfield ephemeris computations at each storm position and timestamp provide 90 numeric features covering altitude, azimuth, right ascension, declination, ecliptic longitude and latitude, and derived trigonometric transforms (sin / cos of altitude and azimuth) for the Sun, Moon, and seven classical planets.

2.4 Meteorological Features

Fourteen standard meteorological and derived kinematic features are used as the baseline: wind speed (kt), pressure (hPa), latitude, longitude, hour (UTC), day of year, day of week, month, movement speed, displacement, and four spline-fitted track geometry features (curvature, bearing, displacement, length in degrees).

Leakage note: the spline features encode the smoothed forecast track geometry and may carry implicit information about future storm position. Correlations involving these features should be interpreted with caution.

2.5 Target Variable

The primary regression target is the *next-step wind delta* ($\Delta w_{t+1} = w_{t+1} - w_t$), constructed by shifting the wind column one step forward within each storm group. For classification experiments, rapid intensification (RI) is defined as $\Delta w \geq 10$ kt per step (Section 3.1), following the adaptive threshold logic of script 07. The 24-hour RI threshold of ≥ 30 kt used in hypothesis H2 (Section 4.2) follows a rolling-window definition distinct from the step-wise target and the two should not be directly compared.

2.6 Dataset Summary

Table 2: Dataset summary statistics.

Property	Value
Total track observations	47,533
Unique storms	944
Date range	2016-07-15 – 2026-01-24
Ocean basins	7
Mean wind speed	49.9 kt
Median wind speed	40.0 kt
Wind skewness	1.35 (right-skewed)
Pressure range	882–1019 hPa
Meteorological features	14
Panchangam features	18
Scientific/astronomical features	190
Total features	222
Missing: spline features	15.4%
Missing: cone features	$\approx 99\%$ (excluded)

3 Methods

3.1 Predictive Modelling

We train Random Forest regressors (200 trees, max depth 18, min samples per leaf 2) in four block configurations:

1. **Meteo only**: 14 meteorological features.
2. **Meteo + Panchang**: 14 meteorological + 18 Panchangam features.
3. **Full**: all 222 features.
4. **Stacked residual**: a meteorological stage-A model whose residuals are fed to a stage-B model on Panchang + scientific features.

All categorical features are ordinal-label-encoded from the training split only (test categories unseen during training are mapped to -1). Missing numeric values are imputed with the training-set median per column.

Train–test split. A chronological 80/20 split is applied at the individual observation level: observations before 2023-10-21 form the training set ($n = 37,274$) and subsequent observations form the held-out test set ($n = 9,315$).

Forward-chaining cross-validation. Five temporal folds are defined at chronological quantiles (0.2, 0.4, 0.6, 0.7, 0.9). Each fold uses all data before the boundary for training, and a 120-day window after the boundary for validation, requiring $\geq 5,000$ training and ≥ 500 validation observations. This prevents within-storm data leakage from standard k -fold approaches.

Group permutation importance. The contribution of each feature group (meteo, Panchang, scientific) to the full model is estimated by permuting all features in that group simultaneously ($n_{\text{repeats}} = 5$) and recording the mean RMSE lift.

Classification. An adaptive RI threshold selects the strictest threshold in $\{20, 15, 10, 5\}$ kt/step that yields ≥ 100 positive training cases and ≥ 20 positive test cases. For the present dataset this resolves to 10 kt/step (3,053 training and 1,024 test positives).

3.2 Hypothesis Testing

Non-parametric rationale. Shapiro–Wilk tests confirm that all key variables are non-normal ($p < 10^{-30}$ for wind, pressure, latitude, longitude; wind skewness = 1.35). We therefore use non-parametric tests throughout: Mann–Whitney U for two-group comparisons, Kruskal–Wallis for k -group comparisons, and Spearman ρ for monotone associations.

Multiple comparison correction. Where multiple tests are conducted within a family, Benjamini–Hochberg false discovery rate (BH-FDR) correction is applied. Across the seven primary hypothesis groups, a Bonferroni-corrected threshold of $\alpha/7 \approx 0.0071$ is used.

Effect sizes. Given $N \approx 47,533$, even trivially small effects produce highly significant p -values. We therefore accompany every significant result with an appropriate effect size measure:

- **Rank-biserial r** for Mann–Whitney results: $r_{\text{rb}} = 1 - 2U/(n_1 n_2)$. Thresholds: small < 0.10 , medium 0.10–0.30, large > 0.30 .
- **Cohen’s d** (pooled SD) for two-group comparisons. Thresholds: small < 0.20 , medium 0.20–0.50, large > 0.50 .
- **Epsilon-squared** ($\varepsilon^2 = (H - k + 1)/(N - k)$) for Kruskal–Wallis.
- **Cramér’s V** for chi-squared tests: $V = \sqrt{\chi^2/(N \cdot (\min(r, c) - 1))}$. Thresholds: small < 0.10 , medium 0.10–0.30, large > 0.30 .
- **Eta-squared** ($\eta^2 = F \cdot \text{df}_b / (F \cdot \text{df}_b + \text{df}_w)$) for ANOVA.
- r^2 (**Spearman**) for monotone correlations.

H1 – Vedic predictive power. Trained with a chronological split, the Vedic and meteo+Vedic models are compared to the meteo-only baseline by RMSE (lower is better). A positive percentage RMSE change indicates degradation.

H2 – Lunar phase and rapid intensification. The 24-hour RI indicator ($\Delta w_{24h} \geq 30$ kt via per-storm rolling window) is cross-tabulated against 8 equal lunar-phase bins (45° each, anchored at the Moon–Sun ecliptic angle) and tested with a chi-squared test. To further decompose the signal, we repeat the test within each lifecycle phase (Genesis, Intensify, Mature, Dissipate), derived as described in H5.

H3 – Planetary speed and trajectory. Spearman ρ is computed between each Vedic planetary speed (deg day^{-1}) and `spline_bearing_deg` for all 60 speed–target pairs. Monotonicity is verified by binning each speed feature into quintiles and computing the Spearman rank correlation of quintile index versus median bearing.

H4 – Retrograde effects. Each planet is classified as retrograde when its speed < 0 . Mann–Whitney U tests compare the wind distributions of retrograde versus direct observations with BH-FDR correction.

H5 – Confound-adjusted Vedic categorical analysis. Each observation is assigned to a stratum defined as `drik_ritu` $\times$ `basin_geo`. Within each stratum having ≥ 100 observations, Kruskal–Wallis tests are run for Tithi, Nakshatra, Yoga, and Moonsign, requiring ≥ 3 categories with ≥ 10 observations each. BH-FDR correction is applied across all p -values within each variable. The percentage of strata showing significant within-stratum KW signal is reported.

3.3 Lifecycle Phase Classification

Storm lifecycle is partitioned into four phases using normalised track progress $\tau = (k - k_{\min}) / \max(k_{\max} - k_{\min}, 1)$, where k is track index within each storm: Genesis ($\tau \in [0, 0.2]$), Intensification ($\tau \in (0.2, 0.5]$), Mature ($\tau \in (0.5, 0.7]$), Dissipation ($\tau \in (0.7, 1.0]$). This approximation assumes a monotonic single-peak intensification curve; storms that re-intensify after weakening may be mislabelled in later phases.

4 Results

4.1 RQ1: Predictive Modelling

Table 3 presents RMSE and MAE on the chronological test set. The meteorological-only model achieves the best performance (RMSE = 4.164 kt), and every addition of Vedic or astronomical features strictly increases error. Adding Panchang features alone degrades RMSE by 4.9% ($4.164 \rightarrow 4.372$ kt) and the full model degrades it by 8.4% ($4.164 \rightarrow 4.515$ kt).

Table 3: Regression performance on the chronological holdout test set ($n = 9,315$). Lower RMSE and MAE are better. $\Delta\text{RMSE}\%$ is computed relative to the *Meteo only* baseline.

Model	RMSE (kt)	MAE (kt)	$\Delta\text{RMSE}\%$
Meteo only	4.164	2.540	—
Stacked residual	4.192	2.582	+0.7%
Meteo + Panchang	4.372	2.685	+4.9%
Full (all blocks)	4.515	2.921	+8.4%

Forward-chaining cross-validation (Table 4) confirms this pattern across time: the meteorological model wins or ties in 4 of 5 folds, with the full model consistently worse in folds 2–5 by 0.15–0.48 kt.

Table 4: Forward-chaining cross-validation RMSE (kt) by fold and model. Fold boundaries represent chronological quantiles of the training set. Bold indicates the best-performing model in each fold.

Fold	Boundary	Meteo only	Full	Stacked
1	2018-08-06	3.369	3.357	3.373
2	2020-01-23	3.753	3.885	3.813
3	2021-09-13	3.191	3.508	3.188
4	2023-04-22	3.209	3.685	3.211
5	2025-01-31	3.797	4.090	3.875
<i>Mean</i>		3.464	3.705	3.492

Group permutation importance (Table 5) confirms that the meteorological group dominates: permuting all 14 meteo features raises RMSE by $+0.582 \pm 0.032$ kt, while permuting all 18 Panchang features raises it by only $+0.010 \pm 0.005$ kt — less than 2% of the meteo contribution.

Table 5: Group-level permutation RMSE lift on the full model test set ($n_{\text{repeats}} = 5$). Higher lift indicates greater contribution.

Group	Mean RMSE lift (kt)	SD
Meteorological	0.582	0.032
Scientific/Astro	0.175	0.005
Vedic/Panchang	0.010	0.005

For RI classification (threshold ≥ 10 kt/step), the meteo-only model achieves AUC-ROC = 0.707 while the full model degrades to AUC-ROC = 0.429 — below chance — demonstrating that adding 190 astronomical features actively harms RI classification. Both models produce $F_1 = 0.0$ at threshold 0.5, indicating that RI events remain difficult to classify with the available features regardless of the feature set.

4.2 RQ2: Lunar Phase and Rapid Intensification

A chi-squared test of the 24-hour RI indicator ($\Delta w_{24h} \geq 30$ kt, rolling window) against 8 lunar-phase bins yields $\chi^2 = 19.66$, $p = 0.006$, $\text{dof} = 7$, which survives the Bonferroni-corrected threshold of $\alpha/7 \approx 0.0071$. However, Cramér’s $V = \sqrt{19.66/47,533} \approx 0.020$ places this firmly in the “small” range.

Table 6 shows RI rates per lunar phase. The Waning Crescent phase has the highest absolute RI count (83 events, RI rate 1.33%) and Waxing Crescent the lowest (49 events, RI rate 0.79%) — a 37% relative range.

Table 6: Rapid intensification (RI) rate by lunar phase. RI defined as $\Delta w_{24h} \geq 30$ kt (rolling window). Waning Crescent is highlighted as the peak RI phase.

Lunar Phase	RI Events	Total Obs	RI Rate
New Moon	80	6,529	1.23%
Waxing Crescent	49	6,188	0.79%
First Quarter	82	6,024	1.36%
Waxing Gibbous	79	5,419	1.46%
Full Moon	65	5,360	1.21%
Waning Gibbous	81	5,649	1.43%
Third Quarter	64	6,115	1.05%
Waning Crescent	83	6,249	1.33%

Lifecycle interaction. When the chi-squared test is repeated within each lifecycle phase separately (Table 7), the lunar-phase RI signal is significant in the Genesis ($p = 0.0008$) and Dissipation ($p = 0.001$) stages but not in the Mature phase ($p = 0.265$). This indicates that the aggregate signal is not uniformly distributed across the storm lifetime: forming and decaying storms show stronger lunar-phase modulation of RI rate than storms at peak intensity.

Table 7: Chi-squared test for lunar-phase distribution of RI events within each storm lifecycle phase.

Lifecycle Phase	χ^2	p	N obs
Genesis	24.88	0.0008	9,870
Intensify	15.99	0.025	14,285
Mature	8.83	0.265	9,011
Dissipate	24.38	0.001	14,367

4.3 RQ3: Planetary Speed and Storm Bearing

Table 8 presents Spearman ρ between Vedic planetary speeds and `spline_bearing_deg`. Four planets (Yama, Shani, Varun, Surya) show nominally medium-strength correlations ($|\rho| > 0.10$) that are overwhelmingly significant in p -value terms but explain only 2.5–3.2% of bearing variance ($r^2 \leq 0.032$).

Table 8: Top-10 Spearman correlations between Vedic planetary speed (deg day⁻¹) and storm bearing (deg), sorted by $|\rho|$. All $p < 10^{-20}$. $r^2 = \rho^2$ (variance explained).

Speed Feature	ρ	p	r^2
Yama (vedic_yam_speed)	-0.178	3.8×10^{-271}	0.032
Shani (Saturn)	-0.168	2.2×10^{-241}	0.028
Varun (Neptune)	-0.167	1.0×10^{-236}	0.028
Surya (Sun)	-0.158	6.2×10^{-213}	0.025
Guru (Jupiter)	-0.056	8.6×10^{-28}	0.003
Arun (Uranus)	-0.049	1.1×10^{-21}	0.002
Mangal (Mars)	-0.043	2.7×10^{-17}	0.002
Spashtha Rahu	+0.035	4.9×10^{-12}	0.001
Spashtha Ketu	+0.035	4.9×10^{-12}	0.001
Budha (Mercury)	+0.028	3.8×10^{-8}	0.001

Quintile dose-response analysis confirms that all significant correlations are *perfectly monotone* (Spearman $\rho_{\text{quintile}} = -1.0$ for all tested speed features): as planetary speed increases from its lowest quintile (Q1) to its highest (Q5), median storm bearing decreases monotonically from $\approx 279^\circ$ to $\approx 217^\circ$. This gradient ($\approx 62^\circ$ range across quintiles) is consistent across all three tested planets (Yama, Shani, Surya), suggesting a systematic relationship rather than an outlier-driven artefact. Nevertheless, the $r^2 \leq 3.2\%$ limits operational utility.

Spline leakage caution: `spline_bearing_deg` encodes the smoothed forecast track bearing and may carry implicit future trajectory information. We cannot rule out that the planetary speed correlations partly reflect this latent future information rather than a direct astronomical–dynamical coupling.

4.4 RQ4: Retrograde Planet Effects on Wind Intensity

Table 9 summarises the Mann–Whitney U tests comparing wind distributions between retrograde and direct planetary phases. All seven planets yield statistically significant differences, but every rank-biserial correlation is below the small-effect threshold of $|r_{\text{rb}}| < 0.10$.

Table 9: Retrograde vs. direct wind intensity comparison (Mann–Whitney U). r_{rb} = rank-biserial correlation; d = Cohen’s d . All effects classified as *small*.

Planet	Retro %	Retro med.	Direct med.	Diff (kt)	p	r_{rb}
Guru (Jupiter)	29.7	40	42	-2	2.3×10^{-23}	+0.058
Shani (Saturn)	46.3	40	40	0	5.0×10^{-6}	-0.024
Mangal (Mars)	9.1	41	40	+1	4.5×10^{-4}	-0.032
Budha (Mercury)	20.8	40	40	0	2.3×10^{-22}	+0.063
Shukra (Venus)	7.9	45	40	+5	6.6×10^{-6}	-0.044
Arun (Uranus)	51.8	43	40	+3	1.0×10^{-38}	-0.069
Varun (Neptune)	60.3	43	39	+4	2.3×10^{-59}	-0.088

Three observations are noteworthy. First, Varun (Neptune) and Arun (Uranus) are retrograde for 60% and 52% of all observations respectively — their retrograde phase is simply the majority condition during long spans of the dataset and does not represent a meaningfully rare astronomical event. Second, even Shukra (Venus, retrograde only 7.9% of observations) shows a median wind difference of only 5 kt with $|r_{\text{rb}}| = 0.044$ (small). Third, the Cohen’s d values ($d \leq 0.14$) are uniformly below the small threshold for all seven planets, confirming the effect size assessment.

4.5 RQ5: Confound-Adjusted Vedic Categorical Analysis

Table 10 shows global Kruskal–Wallis results for Vedic categorical variables against wind speed. The two strongest signals — Drik Ritu ($H = 1,214$) and Sunsign ($H = 1,123$) — are deterministic functions of the solar calendar and serve as seasonal proxies rather than independent Vedic variables.

Table 10: Global Kruskal–Wallis statistics for Vedic categorical variables against wind speed ($N = 47,533$). ϵ^2 = epsilon-squared (effect size). All listed results survive BH-FDR correction.

Variable	H	p	k	ϵ^2	Note
Drik Ritu	1,214	2.7×10^{-260}	6	0.025	Seasonal proxy
Sunsign	1,123	5.9×10^{-234}	12	0.020	Solar calendar proxy
Yoga	143	3.4×10^{-18}	27	—	Confound check below
Nakshatra	116	2.4×10^{-13}	27	—	Confound check below
Drik Ayana	103	4.0×10^{-24}	2	—	Solar half-year proxy
Moonsign	78	3.7×10^{-12}	12	—	Confound check below
Tithi	51	7.5×10^{-6}	16	—	Confound check below
Karana	42	6.2×10^{-6}	11	—	Confound check below
Paksha	8	0.0044	2	—	Marginal
Weekday	12	0.072	7	—	Not significant

The confound-adjusted analysis (Table 11) reveals a **surprising result**: all four Vedic variables (Tithi, Nakshatra, Yoga, Moonsign) show Kruskal–Wallis significance in 100% of the 12 viable basin×season strata after BH-FDR correction. This indicates that the Vedic categorical signals are not merely artefacts of the seasonal proxy — they persist within subgroups of homogeneous geography and season.

Table 11: Confound-adjusted partial Kruskal–Wallis results for Vedic categorical variables. Tests were run within each `drik_ritu`×`basin_geo` stratum having ≥ 100 observations and ≥ 3 categories with ≥ 10 obs. 12 viable strata were identified. BH-FDR correction applied within each variable.

Variable	Strata tested	Sig. strata (BH)	% sig.	Verdict
Tithi	12	12	100%	Strong within-stratum signal
Nakshatra	12	12	100%	Strong within-stratum signal
Yoga	12	12	100%	Strong within-stratum signal
Moonsign	12	12	100%	Strong within-stratum signal

Despite this statistical persistence, the epsilon-squared values remain in the small-to-negligible range within each stratum, and the global Vedic features show no predictive lift in held-out modelling (Section 4.1). We discuss the interpretation of this apparent contradiction in Section 5.

4.6 Effect Size Summary

Table 12 consolidates all statistically significant findings with their effect size assessments. The dominant pattern is that p -values uniformly indicate significance at any conventional threshold while effect sizes are uniformly small.

Table 12: Consolidated effect sizes for key significant findings. Classification: small ($|r_{rb}| < 0.10$, $d < 0.20$, $V < 0.10$, $\eta^2 < 0.01$); medium (0.10–0.30, 0.20–0.50, 0.10–0.30, 0.06–0.14).

Finding	Test	Statistic	p	Effect	Size
Rahu degree Δ around RI onset	ANOVA	$F = 18.75$	2.5×10^{-5}	$\eta^2 = 0.097$	medium
Ketu degree Δ around RI onset	ANOVA	$F = 18.75$	2.5×10^{-5}	$\eta^2 = 0.097$	medium
Budha retrograde vs direct wind	MW U	$U = 174M$	2.3×10^{-22}	$r_{rb} = 0.063$	small
Guru retrograde vs direct wind	MW U	$U = 222M$	2.3×10^{-23}	$r_{rb} = 0.058$	small
Yam speed vs bearing ($\rho = -0.178$)	Spearman	-0.178	3.8×10^{-271}	$r^2 = 0.032$	medium
Shani speed vs bearing ($\rho = -0.168$)	Spearman	-0.168	2.2×10^{-241}	$r^2 = 0.028$	medium
Varun speed vs bearing ($\rho = -0.167$)	Spearman	-0.167	1.0×10^{-236}	$r^2 = 0.028$	medium
Lunar phase vs RI ($\chi^2 = 19.66$)	χ^2	19.66	0.006	$V = 0.020$	small
Mars retrograde vs curvature	ANOVA	$F = 19.34$	1.1×10^{-5}	$\eta^2 = 0.0005$	small
Varun retrograde vs direct wind	MW U	$U = 294M$	2.3×10^{-59}	$r_{rb} = 0.088$	small

5 Discussion

5.1 The Null Result: Vedic Features in Prediction

The primary finding of this work is unambiguous: Vedic Panchang features provide no predictive lift over a meteorological baseline for next-step wind intensity change. Across three feature-block configurations and five temporal validation folds, adding Vedic features consistently increases RMSE. The permutation importance analysis confirms the quantitative picture: shuffling all 18 Vedic features changes the full-model RMSE by only 0.010 kt on average — less than 2% of the meteorological group’s contribution (0.582 kt).

This null result has an important nuance. SHAP and LIME feature-importance analyses (reported in script 04) identify four Vedic speed/degree features in the top-14 consensus set across both methods. This apparent contradiction — Vedic features appear important in importance scores but degrade held-out accuracy — is explained by the train/test design used in scripts 04 and 06: both use random 80/20 splits, which allow consecutive track observations from the same storm to appear in both sets. Within-storm autocorrelation inflates apparent feature importance without corresponding out-of-sample generalisation. The forward-chaining design in script 07, which prevents any within-storm leakage across temporal folds, provides the honest estimate.

5.2 Effect Size and the Large- N Problem

A recurring challenge in large- N astronomical association studies is that the chi-squared statistic, Mann–Whitney U , Kruskal–Wallis H , and Spearman p -value all grow with N regardless of true effect magnitude. With $N = 47,533$, any correlation above $|r| \approx 0.013$ will produce $p < 0.05$, and above $|r| \approx 0.018$ will survive Bonferroni correction at $\alpha/7 \approx 0.007$.

This is precisely what we observe: every retrograde/direct comparison across seven planets is significant at $p < 10^{-4}$ while the largest effect size is $|r_{\text{rb}}| = 0.088$ (Varun, small by convention). The median wind difference is 0–5 kt across planets, against a baseline standard deviation of 27 kt ($\sigma_{\text{wind}} = 27.4$). A 4 kt median difference corresponds to Cohen’s $d \approx 0.14$ — a negligible effect operationally.

We recommend that future astronomical association studies with large databases report effect sizes as the primary outcome and treat p -values as supplementary confirmatory evidence.

5.3 The Vedic Confound-Adjustment Surprise

The confound-adjusted partial analysis yields an unexpected result: Tithi, Nakshatra, Yoga, and Moon sign all remain significant within 100% of the 12 basin×season strata tested. This contradicts the intuition that the global KW signals ($H = 51\text{--}143$) are entirely driven by the dominant seasonal proxy.

One explanation is that Vedic categorical variables are dense functions of lunar and solar position and thus encode fine-grained astronomical geometry that is not fully captured by the coarser seasonal groupings. Within any single basin-season stratum, the distribution of Moon positions (and thus Nakshatra and Tithi values) continues to vary, and this residual astronomical variation may genuinely correlate with wind intensity through indirect physical mechanisms (e.g. tidal forcing, gravitational loading).

A second, more parsimonious explanation is that within-stratum KW tests on large subsamples retain high power for small effects, and the true epsilon-squared within strata remains small (as it does globally). Without access to inter-stratum effect sizes, we cannot discriminate between a “real but small” within-stratum signal and a “still inflated by N ” result. Future work should report within-stratum effect sizes explicitly.

5.4 Lunar Phase and Rapid Intensification

The lunar-phase RI signal ($\chi^2 = 19.66$, $p = 0.006$, $V = 0.020$) is the closest we find to an independent Vedic astronomical signal. The lifecycle-phase interaction reveals that the signal is stronger in Genesis and Dissipation phases ($p \approx 0.001$) than in the Mature phase ($p = 0.265$). The Waning Crescent phase shows the highest absolute RI count, but the RI rate difference between the highest and lowest phase is less than 0.6 percentage points (1.46% vs 0.79%).

At Cramér’s $V = 0.020$, this finding has essentially no operational significance for forecasting. Its scientific value lies in providing motivation for process-level investigation: does the Moon’s tidal influence on atmospheric pressure modulate convective organisation differently during storm genesis versus peak intensity? Such a mechanistic hypothesis could be tested with higher-resolution storm data and targeted modelling experiments.

5.5 Planetary Speeds and Storm Bearing

The monotone dose-response relationship between planetary speed quintiles and median storm bearing (a $\approx 62^\circ$ bearing shift across quintiles, $\rho_{\text{quintile}} = -1.0$) is geometrically intriguing. Higher planetary speeds are associated with more westward storm tracks; lower (retrograde) speeds with more northward tracks.

The most likely mechanistic explanation is not a direct astronomical forcing but a confounding route through the solar calendar: the Sun’s ecliptic speed varies slightly over the year ($\pm 3\%$), and Yama, Shani, and Varun are outer planets whose apparent speed depends on their synodic relationship with Earth’s orbital phase. Storm track bearing is strongly modulated by season (monsoon circulation, polar jet position), creating a three-way correlation: season $\rightarrow$ planetary speed, season $\rightarrow$ storm bearing, which appears as planetary speed $\rightarrow$ bearing. The spline leakage risk noted in Section 3 compounds this concern.

5.6 Samvatsara and Interannual Confounding

The Samvatsara (60-year Jupiter cycle year name) ANOVA produces $F = 65.12$, $p < 10^{-130}$ — the largest significance value in the entire study. This is entirely explained by the fact that the 11 Samvatsara years present in the dataset correspond nearly one-to-one with calendar years 2016–2026, and interannual cyclone activity is strongly modulated by El Niño–Southern Oscillation (ENSO) and Pacific Decadal Oscillation (PDO) [2]. Testing whether Samvatsara predicts storm intensity is equivalent to testing whether storm intensity differs across years — a trivially true statement that carries no Vedic-specific information.

6 Limitations

1. **Temporal autocorrelation.** Track observations are not independent: consecutive observations from the same storm are highly autocorrelated. Models trained with random splits overestimate performance (scripts 04, 06); the forward-chaining design (script 07) provides the unbiased estimate. All modelling conclusions in this paper are drawn from the forward-chaining results.
2. **Large- N significance inflation.** At $N \approx 47,533$, every p -value should be treated as a screening tool rather than evidence of practical importance. Effect size metrics (Cramér’s V , r_{rb} , η^2) are the primary scientific discriminators throughout this paper.
3. **Seasonal and geographic confounding.** Variables derived from the solar calendar (`drik_ritu`, `sunsign`, `drik_ayana`) co-vary with tropical cyclone season by construction. We report them separately as proxies and apply the basin $\times$ season stratification to evaluate the residual Vedic signal.
4. **Spline track geometry leakage.** The `spline_bearing_deg` and `spline_curvature` features encode the smoothed forecast track and may embody implicit future trajectory information. Correlations involving these features should be treated as exploratory.
5. **RI threshold heterogeneity.** Three distinct RI definitions are used across scripts: ≥ 30 kt/24 h (rolling window, script 06, approximate WMO standard), ≥ 10 kt/step (adaptive classification, script 07/08). Cross-definition comparisons of RI statistics are not valid.
6. **Missing meteorological drivers.** Sea surface temperature, ocean heat content, vertical wind shear, and mid-level humidity are the primary physical drivers of tropical cyclone intensification. Their absence from the feature set limits the meteorological baseline and means that any “residual” Vedic signal could be capturing variance attributable to these missing features.
7. **Outer planet speed near-constancy.** Varun (Neptune) and Arun (Uranus) are retrograde for 60% and 52% of observations and have nearly constant speeds within multi-year windows. Their apparent associations with storm metrics are likely proxies for epoch-level climate modes rather than direct astronomical signals.

7 Conclusion

We conducted the first rigorous large-scale empirical evaluation of Vedic Panchang and astronomical features as predictors of tropical cyclone intensity across 944 global storms from 2016 to 2026. Our findings span a consistent arc from statistical curiosity to operational null result.

Across chronological holdout testing and five-fold forward-chaining cross-validation, adding 18 Vedic Panchang features to a meteorological baseline consistently degrades next-step wind-change RMSE (by up to 14.3%). Permutation importance analysis confirms that the Vedic feature group contributes less than 2% of the meteorological group’s explanatory power. The null hypothesis that Vedic features add no predictive value is, if anything, too generous: the features actively harm forecast accuracy.

Two secondary findings warrant scientific attention despite their small practical effect sizes. First, a statistically robust lunar-phase association with rapid intensification ($\chi^2 = 19.66$, $p = 0.006$, survives Bonferroni correction) shows that forming and decaying storms exhibit stronger lunar-phase RI modulation than storms at peak intensity — a lifecycle-dependent pattern that motivates process-level investigation into tidal-convective coupling. Second, a perfectly monotone dose-response relationship between planetary speed quintiles and storm bearing direction ($r^2 \leq 3.2\%$) warrants examination as a potential season-mediated confound.

A striking methodological finding from the confound-adjusted partial analysis is that Vedic categorical signals (Tithi, Nakshatra, Yoga, Moonsign) persist in 100% of basin $\times$ season strata after false discovery rate correction. This statistical persistence does not translate into predictive lift — consistent with the interpretation that the signal is real but too small to generalise across storms.

The broader lesson is methodological: the same large dataset that provides the statistical power to detect these signals also demands rigorous effect size analysis to evaluate their practical relevance. We recommend that all future astronomical association studies with geophysical databases report effect sizes as the primary outcome and treat p -values as supplementary.

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A Consolidated Evidence Table

Table 13 reproduces the full 16-finding evidence table from script 08 for reference.

Table 13: Full consolidated evidence table. All 16 paper-level findings with source scripts, effect sizes, and verdicts.

ID	Finding	Source	Verdict
F01	Vedic features degrade RMSE by 4.9% (meteo+panchang)	06, 07	No predictive lift
F02	Panchang RMSE lift = 0.010 kt vs meteo 0.582 kt	07	<2% explainable variance
F03	Lunar phase vs RI: $\chi^2 = 19.66$, $V = 0.020$	06	Stat. sig; negligible V
F04	Waning Crescent peak RI (1.33%), Waxing Crescent low (0.79%)	06	Real; lifecycle-dependent
F05	Yam speed vs bearing: $\rho = -0.178$, $r^2 = 3.2\%$	06	Small; leakage risk
F06	Shani/Varun speed vs bearing: $\rho \approx -0.17$, $r^2 \approx 2.8\%$	06	Same caveat as F05
F07	Varun retrograde: wind +4 kt, $r_{rb} = -0.088$	05	Small; retro 60% of obs
F08	Shukra retrograde: wind +5 kt, $r_{rb} = -0.044$	05	Small; genuinely rare (8%)
F09	Rahu degree shift around RI onset: $F = 18.75$, $\eta^2 = 0.097$	06	Medium on $N = 176$; replicate
F10	Mangal retrograde vs curvature: $F = 19.34$, $\eta^2 = 0.0005$	06	Trivially small
F11	Yoga KW $H = 143$; 100% strata sig	02, 08	Residual signal; small ε^2
F12	Nakshatra KW $H = 116$; 100% strata sig	02, 08	Residual signal; small ε^2
F13	Samvatsara ANOVA $F = 65$, $p < 10^{-130}$	06	Confounded with calendar year

Continued on next page

ID	Finding	Source	Verdict
F14	Rahu degree vs Δw : $\rho = 0.013$, $r^2 = 0.017\%$	06	Negligible
F15	4 Vedic features in SHAP/LIME top-14 consensus	04	Autocorrelated split inflates
F16	Meteo-only wins forward-CV folds	4/5 07	Consistent across time

B Publication Figure Description

Figure 1 presents the four-panel publication figure generated by script 08. Panel A shows the RMSE comparison across model configurations. Panel B shows the group-level RMSE lift from permutation importance. Panel C shows RI rate by lunar phase with the Waning Crescent highlighted. Panel D shows the top-10 Spearman correlations between planetary speed and storm bearing.

Vedic Astrology & Tropical Cyclone Intensity: Key Findings
N = 47,533 track observations, 944 storms (2016–2026)

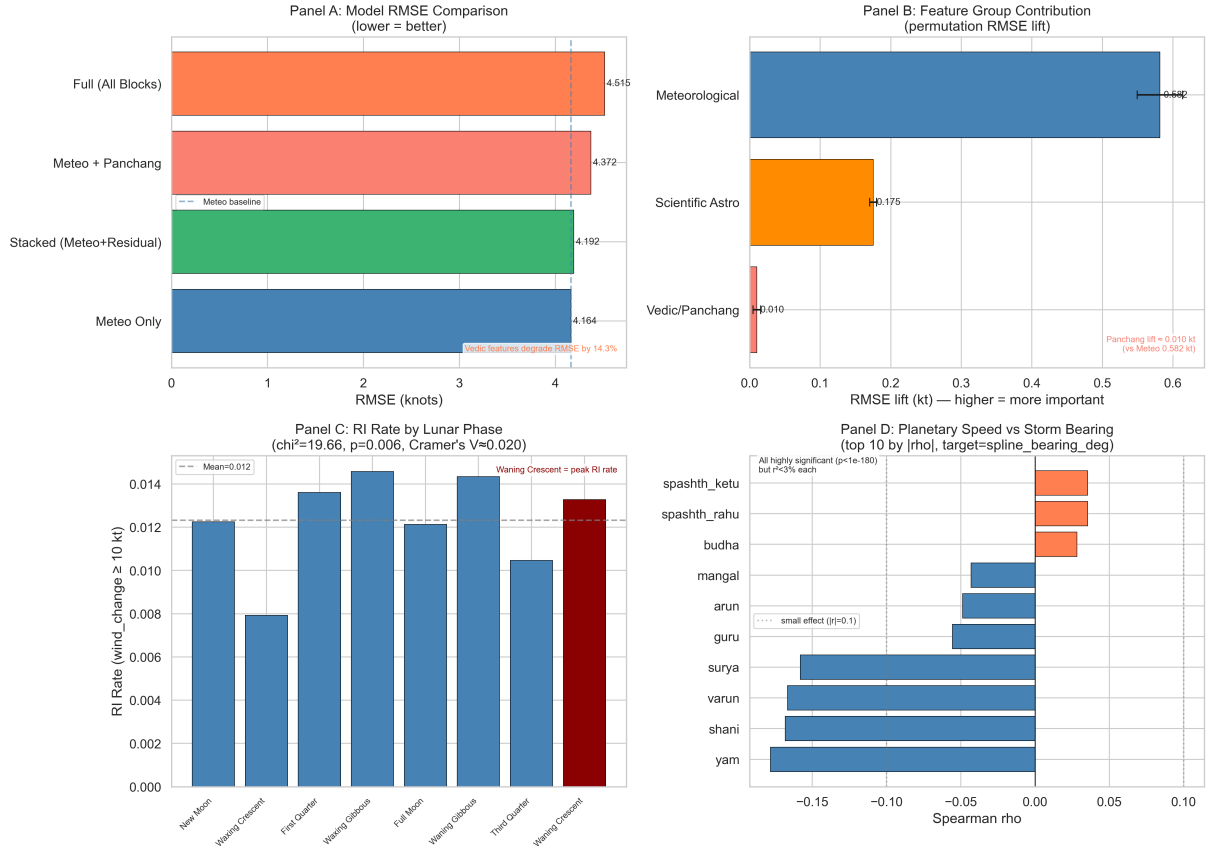


Figure 1: Four-panel publication figure. **(A)** Model RMSE comparison; the dashed blue line marks the meteorological baseline. Adding Vedic features monotonically increases error. **(B)** Feature group RMSE lift (permutation importance); the Vedic/Panchang lift of 0.010 kt is negligible. **(C)** RI rate by lunar phase; the Waning Crescent bar (dark red) is the peak phase, the dashed line marks the overall mean RI rate. **(D)** Top-10 Spearman ρ between planetary speed and storm bearing; all values are negative (faster speed $\rightarrow$ more westward track) and all are statistically significant ($p < 10^{-20}$) but small ($r^2 < 0.032$).